

Northern University of Business and Technology Khulna

Department of Computer Science and Engineering

Week Wise Project Breakdown

Course Title: Software Development I

Course Code: CSE2216

Section: 4B

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Date of Submission: 21.06.2026

The name of the project I'm about to do is **Advanced Stopwatch with Additional Features**.

The watch will contain features like,

1. **Stopwatch**
2. **Time calculating**
3. **Time freezing**

For the stopwatch option there will be **Start, Pause, Resume, Reset button**. For calculating the time there will be options for selecting specific time and date. Such as **Years, Months, Days, Hours, Minutes** and **Seconds**. And for the freezing part it will be like time freeze for 2 minutes(example).

Hence this is quite a large project, I've decided to breakdown the whole thing into 10 weeks. For each particular week, there will be individual tasks. The week wise tasks are broken down below:

Week 1:

1. Planning about what features the watch will contain.
2. The language and materials that will be used to build the watch.
3. Collection of materials such as a working computer, JDK, a compiler to construct and run the code.
4. Making a whole report on the project.

Week 2:

1. Draw UI sketch.
2. Create use case diagram.
3. Create class diagram.

Week 3:

1. Main window.
2. Time display label.
3. Start button.
4. Pause button.
5. Reset button.

Week 4:

A fully functional stopwatch with the following features:

1. Start.
2. Pause.
3. Reset.

Week 5:

Auditioning the following features to the watch

1. Resume.
2. Millisecond precision.

Week 6:

Week 6 is basically adding the time and date difference calculator to the project.

User enters date. Example:

01-01-2020(for date)

Program calculates:

6 years
5 months
20 days

Use:
Local Date
Period
Duration

Week 7:

On week 7 the target is to design Freeze Time Feature. For example:

Freeze time from:

Freeze time to:

Week 8:

The task of this week is to add features for the freezing information to get stored. For this we need to create freezeData.data. This is a simple file storage. This won't need any database. For example:

When the user freezes the stopwatch:

Freeze Until:
2026-12-25 10:00 AM

Elapsed Time:
8100000 ms

The program can simply write that information to a file.

Week 9:

The task of this week is to add the part for the unfreeze. For this we need to do the following steps,

On startup:

loadState()

Check:

now > freezeUntil

If true: unfreeze()

Otherwise: remain frozen

Week 10:

The task for this week is Testing and Bug Fixing.

The Stopwatch

1. Start.
2. Pause.
3. Reset.
4. Resume.

Freeze

1. Freeze for.
2. Exit program.
3. Reopen.

Date Calculator

1. Past date.
2. Future date.
3. Invalid date.

If the outcome of this project doesn't arrive like this, then there must be bugs and we need to fix those. And if it arrives including all these options then we have to understand that the project is successful and if we want we can also publish it hence it is a **GUI** based application and has enough capability to satisfy the user.

